ABDULLAH SAIF

Mechanical Engineering Undergraduate | CAD • Embedded Systems • Data Analysis

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PROFILE SUMMARY

Mechanical Engineering undergraduate at UET Lahore with hands-on experience in CAD design, 3D modeling, embedded systems, and data analysis. Proficient in AutoCAD (Certified), Python (NumPy, Pandas, Matplotlib), and Arduino/ESP32 platforms. Skilled in IoT prototyping, sensor integration, technical documentation, and data-driven problem solving. Seeking internship opportunities in mechanical engineering, CAD drafting, manufacturing, embedded systems, or data analysis to apply academic and project-based expertise in industry settings.

CORE COMPETENCIES

CAD Design & 3D Modeling • AutoCAD (Certified) • SolidWorks • Engineering Drawing & GD&T • 2D Drafting

Embedded Systems • Arduino Programming • ESP32 • IoT Prototyping • Sensor Integration • Microcontroller Programming

Data Analysis • Data Visualization • Python (NumPy, Pandas, Matplotlib) • Technical Documentation • MATLAB

Fluid Mechanics • Thermodynamics • Mechanics of Materials • Force & Load Analysis • Prototyping • Simulation

TECHNICAL SKILLS

CAD Software: AutoCAD (Certified), SolidWorks (learning), Engineering Drawing, GD&T, 2D Drafting, 3D Modeling

Programming: Python, Arduino C/C++

Libraries & Tools: NumPy, Pandas, Matplotlib, MATLAB, Latex

Embedded Systems: ESP32, Arduino, IoT Prototyping, Sensor Integration, Microcontroller Programming, Data Acquisition

Engineering: Fluid Mechanics, Thermodynamics, Mechanics of Materials, Force & Load Analysis, Lathe Operations

Documentation: Technical Report Writing, Engineering Presentation Delivery, Latex documentation

Languages: English (Proficient), Urdu (Fluent), Punjabi (Native)

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern | Nishat Mills Limited – Dyeing & Finishing Plant, Lahore *Jul – Aug 2026*

- Completed a 4-week hands-on internship in the Mechanical Department, rotating through workshop stations covering AC systems, welding, lathe machining, and woodworking
- Studied plant utility systems including vapor-compression chillers, the water softening plant, and an Atlas Copco GA55 rotary screw air compressor
- Assisted the maintenance team with inspection and servicing of bearings, pumps, valves, gears, and belts across production machinery
- Applied first-principles thermodynamic and fluid mechanics analysis to real industrial equipment, strengthening the link between classroom theory and plant-level engineering practice

EDUCATION

BSc Mechanical Engineering | University of Engineering & Technology (UET) Lahore *2024 – Present*

- Expected Graduation: 2028 | CGPA: 3.2 | Completed: 4th Semester
- Relevant Coursework: Fluid Mechanics, Thermodynamics, Mechanics of Materials, Engineering Mechanics, Machine Design, CAD, Engineering Programming

FSc Pre-Engineering | KIPS College, Kasur *2022 – 2024*

- Marks: 85% strong foundation in Mathematics and Physics

Matriculation (Science) | District Public School (DPS), Kasur *2020 – 2022*

- Marks: 97% top academic performer

PROJECTS & EXPERIENCE

Hydroponic Room Monitoring and Control System | Embedded Systems Project *2026*

- Designed and built a dual-ESP32 based IoT system for automated hydroponic grow-room monitoring and control
- Interfaced pH, TDS, temperature, humidity, and water-level sensors for real-time data logging
- Integrated Firebase Realtime Database with SD-card logging for redundant, real-time data storage
- Implemented relay-based control of pumps, fans, and RTC-scheduled grow lights; built a web dashboard for live monitoring
- Iteratively debugged firmware across multiple versions, resolving sensor calibration, signal conditioning, and embedded system issues
- Project live at : Thermodynamics-Lab.web.app

ESP32-Based Viscometer | Embedded Systems Project *2025*

- Designed and built a functional viscometer using ESP32 microcontroller; implemented sensor integration and real-time data acquisition to measure fluid viscosity
- Programmed sensor calibration and data processing logic in Arduino C/C++; validated readings against known fluid standards improving measurement accuracy
- Designed complete embedded system architecture; documented test methodology, performance analysis, and system validation in a formal technical report
- Project repository: github.com/Abdullah-saif-18

Automobile Air Filter | CAD Design & 3D Modeling Project *2025*

- Produced accurate 2D drafting and 3D modeling of an automobile air filter using AutoCAD; applied GD&T annotations, dimensioning, and engineering drawing standards
- Delivered complete drawing package suitable for manufacturing reference and product design review

Jib Crane Model & Structural Force Analysis | Engineering Mechanics Project *2024*

- Designed a scaled jib crane model and performed complete static force and load analysis under multiple load conditions; calculated reaction forces, member loads, and factor of safety
- Validated structural simulation results against theoretical models; prepared formal technical documentation and delivered structured presentation to faculty evaluators

Python Data Analysis & Visualization | Programming Project *2025*

- Analyzed engineering datasets using Python: Pandas for data wrangling, NumPy for numerical computation, Matplotlib for data visualization
- Applied data cleaning, filtering, and statistical summary techniques; generated visualizations communicating trends for technical reporting

CAD 3D models | 3D Modeling

- 3D model of Engine Blower
- 3D model of Crane Hook

CERTIFICATIONS

AutoCAD with Basic 3D Modeling | UET Lahore *Jan - Feb 2025*

- Professional certification covering 2D drafting, 3D modeling, CAD drafting standards, and engineering drawing core skills for mechanical design and manufacturing roles

Python Programming Certificate | Google / Coursera *2025*

- Google-certified course covering data structures, Python libraries (NumPy, Pandas), and applied programming for data analysis and automation

MATLAB Onramp | MathWorks Training Services Aug 2026

- Completed self-paced MathWorks training covering MATLAB fundamentals variables, matrices, plotting, and scripting with a 100% completion score

INTERESTS & AREAS OF FOCUS

Industrial Machinery & Maintenance • Robotics & Automation • Embedded Systems & IoT • CAD Design & Manufacturing • Prototyping • Emerging Technologies in Mechanical Engineering